Prediction of Employee Attrition

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Abstract—Employee attrition is basically downfall in an organization where employees depart. They are the most important bodies of an organization who contributes in the growth of organization. In an organization, they are regarded as its core. Employee performance is a key factor in determining an organization's overall performance. Therefore, it's necessary to find out whether or not employees are content and whether they have any additional motivations for quitting.

Based on research, different supervised machine learning algo- rithms such as Logistic Regression, Decision Tree, Naive Bayes Classifier, Random Forest Classifier, and SVM are compared. Comparatively, Random Forest algorithm gives better accuracy on employee attrition dataset. As a result, Random Forest Clas- sification method is used in the research for predicting employee attrition. Here the Machine Learning algorithm uses the Kaggle employee attrition dataset. It includes 35 attributes for 1470 employees such as employee performance, average monthly hours at work, Salary, Job satisfaction, Years in company and others as features. As observed, Random Forest makes predictions with approximately 87.46% accuracy.

Index Terms—Random Forest, Machine Learning, Attrition, Support Vector Machine

I. INTRODUCTION

The most important element of any organization is its employees. They determine whether an organization succeeds or fails. It is essential to figure out whether or not they are satisfied with their jobs. Observations indicate that a lack of Employee attrition is a result of poor self-esteem, overtime pay, employee recognition, and monthly revenue. Employee Attrition causes organizations to hire new employees to con- tinue with their work. New hire will take money as well as time to make the respective organization profitable. When experi- enced employees leave, it is very difficult to hire the employee with same skill, expertise, knowledge and experience. Innova- tion and organizational effectiveness get impacted. Apart from this, employee attrition affects the organizational reputation. Thus, attrition can have adverse impacts on organizations, such as higher hiring expenses, a decline in institutional expertise, lower productivity, and workflow disruptions.

However, Employee Attrition Prediction is able to address each of the problems mentioned above. It helps Human resource (HR) Managers to predict the total number of employees who will leave the organization within a given time frame. By addressing factors contributing to attrition, Managers can work on them and try to retain them. Accurate employee attrition

prediction facilitates long-term workforce planning and talent management strategies. Organizations can anticipate staffing needs, succession planning requirements, and skill gaps, to ensure continuity in their workforce.

The ability to predict and prevent employee attrition is crucial for maintaining a stable and engaged workforce. By under- standing the underlying factors that contribute to attrition, or- ganizations can develop proactive strategies to retain valuable employees and mitigate the risk of turnover [1]. Conventional method to employee retention often rely on reactive strategies, such as exit interviews and employee satisfaction surveys. It may not always provide timely or actionable insights. In contrast, predictive modeling offers a data-driven approach to anticipate attrition trends and identify at-risk employees before they decide to leave.

The research by Akinode [2], on Employees Attrition Prediction Using Machine Learning Algorithm 84% accuracy is obtained for Random Forest Algorithm. In the research comparison is made between different Machine Learning Algorithm so as to find out which one gives the best prediction. Research by Raza et al. [3] gives accuracy of 83%.

Research by Pratt et al. [4] shows the Random Forest with 85.2% accuracy.

Mohbey [5] gives a comparison of several supervised machine learning techniques for employee attrition prediction. The research finds that the Random Forest method yields the greatest accuracy, followed by Logistic Regression.

Nurhindarto et al. employed random forest and univariate ROC feature selection to predict employee attrition and performance. In the comparison between Random Forest and Decision Tree [6], Random Forest is found to be more accurate. The accuracy achieved by Random Forest is 79.16%.

Other than above mentioned research, their are multiple re- search [7], [8], [9], [10], in which different supervised machine learning algorithms such as Logistic Regression, Decision Tree, Naive Bayes Classifier, Random Forest Classifier, and SVM are compared. Comparatively, Random Forest algorithm gives better accuracy.

II. PROPOSED ARCHITECTURE

In the proposed work an attempt is made to predict the employee attrition. For prediction, Random Forest Algorithm is used which is a supervised Machine Learning algorithm [11]. The proposed work has an objective as follows:.

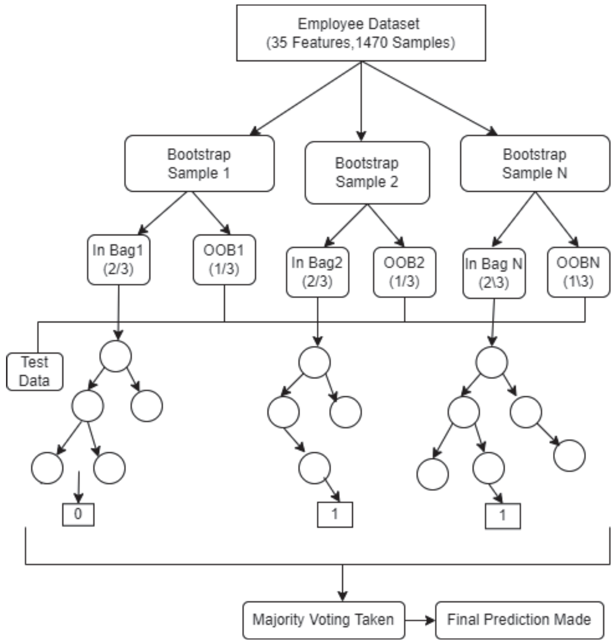


Fig. 1. Random Forest Diagram

- Finding the common reasons of employee spontaneously leaving the organization.
- Identification of the factors which make employees un-satisfy with the job.
- Helps in reduction of employee attrition by suggesting strategies and steps.
- Determining the organizational environment.
- Optimizing the retention process to ensure that we always keep the RIGHT people at RIGHT role at RIGHT time.
- Determining the satisfactory level of employees towards their job and working conditions.
- Accessing the level of employee satisfaction with existing retention strategies.

Python programming language [12] is used for prediction. Random Forest [13] is an ensemble learning approach where multiple Decision Tree model are created in terms of estimator and it takes the combine output as shown in Fig. 1. The divide and conquer strategy is the foundation of ensemble approaches, which aim to increase performance. A number of metrics including accuracy, precision, recall, and F1-score, can be used for evaluation of Random Forest models. These metrics can be found out using confusion matrix. A confusion matrix is a matrix that provides an overview of the performance of a machine learning model on a particular set of test data. Based on the model's predictions, it is a way to show the percentage of accurate and inaccurate instances.

The research involves the following steps [14]:

a) Collection of Employee Dataset.

b) Data preprocessing and exploratory data analysis.

c) Feature engineering [15] for creating new features and relationships out of our existing features.

d) Splitting the dataset into training and test datasets.

e) Training the model on 55% data and predicting the output on 45% data.

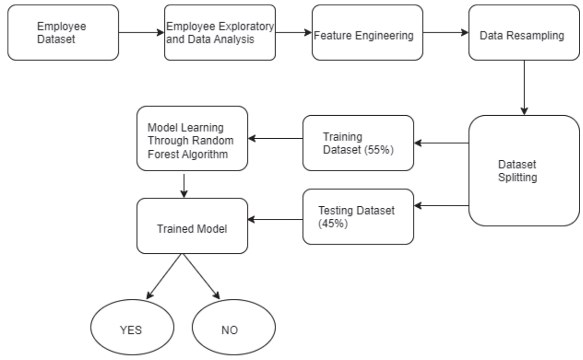


Fig. 2. Architecture Diagram

TABLE I. IBM DATASET FEATURE

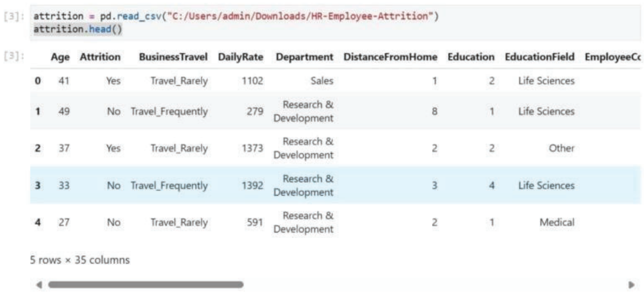
Feature Name	Data Type	Feature Name	Data Type
Age	Numerical	MaritalStatus	Categorical
Business Travel	Categorical	monthlyRate	Numerical
DailyRate	Numerical	NumCompaniesWorked	Numerical
Department	Categorical	Over18	Categorical
DistanceFromHome	Numerical	OverTime	Categorical
Education	Numerical	PercentSalaryHike	Numerical
EducationField	Categorical	PerformanceRating	Numerical
EmployeeCount	Numerical	RelationshipSatisfaction	Numerical
EmployeeNumber	Numerical	StandardHours	Numerical
EnvironmentSatisfaction	Numerical	StockOptionLevel	Numerical
Gender	Categorical	TotalWorkingYears	Numerical
HourlyRate	Numerical	TrainingTimesLastYear	Numerical
JobInvolvement	Numerical	WorkLifeBalance	Numerical
JobLevel	Numerical	YearsAtCompany	Numerical
JobRole	Categorical	YearsInCurrentRole	Numerical
JobSatisfaction	Numerical	YearsSinceLastPromotion	Numerical
MonthlyIncome	Numerical	YearsWithCurrManager	Numerical

B. Employee Dataset Collection

The dataset used for the research is downloaded from kaggle [16]. The data set consists of 35 columns (features) and 1470 rows (instances). It contains both categorical and numerical attributes as shown in TABLE I. Here, Attrition feature (column) represents employee decision whether they will leave the organization or not. Yes indicates that the employee will quit the organization and No indicates that the employee won't be leaving the organization.

C. Data preprocessing and exploratory data analysis

In the research, the data set is downloaded from kaggle and is uploaded to Jupiter notebook in anaconda. The read csv() function from pandas [17] is used to read the CSV file. After exploring dataset, the next step is preprocessing. It includes feature engineering and numerically encoding categorical values. Finding the correlation matrix is the next step in a data exploration. By plotting a correlation matrix, an idea of how features are related to each other is obtained. To do this Plotly library [18] is used.



```
[3]: attrition = pd.read_csv("C:/Users/admin/Downloads/HR-Employee-Attrition")
     attrition.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeC
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	

5 rows × 35 columns

Fig. 3. Overview of imported dataset

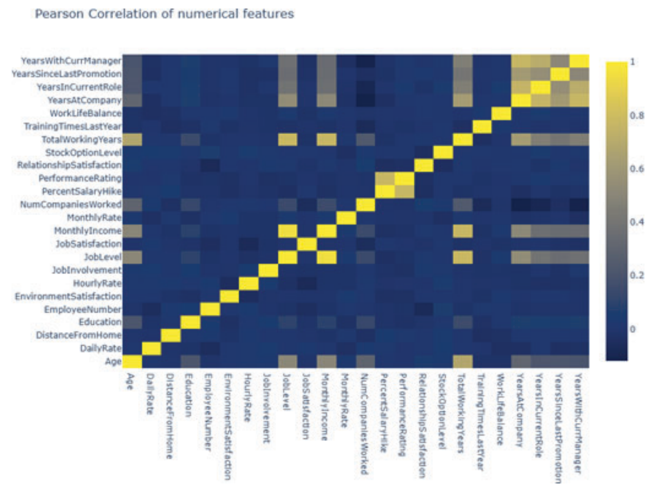


Fig. 4. Correlation

D. Data Resampling

In the present instance, the Attrition column provides a target. As it includes category variables, numeric coding is required. By creating a dictionary with the provided mapping of 1: Yes and 0: No, we digitally encode it.

E. Dataset splitting

Data must be split into training and test datasets before applying machine learning to predict employees attrition. The Random Forest model is evaluated using 45% of the dataset in this instance, whereas the model is trained using 55% of the dataset.

III. Implementation Of Proposed Model

The first step of the research is to load the dataset and perform exploratory data analysis to comprehand the structure of the data.Then, data preprocessing and exploratory analysis is done to obtain a general understanding of the distribution of features. To perform this, kdeplot function in the seaborn library is used. Finding the correlation [19] matrix is the next step in a data exploration process, and it is accomplished using the plotly library. By plotting a correlation matrix as shown in Fig. 4, we get a transparency of how the features are related to each other. So, correlation matrix is formed between all the numerical features. The next step is to carry out the feature engineering and numerical encoding of the categorical values in our dataset after exploring it. Feature engineering is the process of extending our existing features and their relationships to create new features. The first step in digitally encoding the data is determining which of the features include categorical data. Next, data points from the existing dataset are randomly selected for resampling, which creates new data points for the dataset. It facilitates the creation of new synthetic data sets for machine learning model training as well as the estimation of a dataset's characteristics. After this, dataset is distributed into training and test data. For training 55% dataset and for testing 45% data is used.

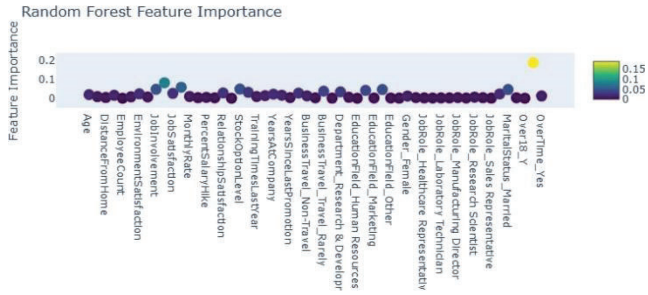


Fig. 5. Feature Importance

A. Algorithms

Algorithm 1 Random Forest Algorithm

Require: Training dataset X, testing dataset X_{test}, actual target values y_{true}
Ensure: Accuracy of the Random Forest model

Split the dataset into training and testing sets (55-45 split).

Set parameters for Random Forest:
  $n\_estimators = 1000$
  $max\_features = 0.3 \times$ total_features {Assuming 35 total features}
  $max\_depth = 4$
Initialize an empty list to store decision trees.
for i from 1 to $n\_estimators$ **do**
  Randomly select $max\_features$ from the total features.
  Train a decision tree with max_depth of 4 using the selected features.
  Add the trained decision tree to the list of decision trees.
end for
Define *Random_Forest_Predict*(X_{test}):
  Initialize an empty list to store predictions.
for each decision tree in the list **do**

  Make predictions using the decision tree on X_{test}.
  Add the predictions to the list of predictions.
end for
Return the majority vote of predictions (i.e., mode) for each sample in X_{test}.
Define *Calculate_Accuracy*(y_{true}, y_{pred}):
  Compare y_{true} (actual target values) with y_{pred} (predicted target values).
  Calculate the accuracy as the ratio of the number of correct predictions to the total number of predictions.
  Return the accuracy.
Call *Random_Forest_Predict*(X_{test}) to predict the target variable for the testing dataset.
Call *Calculate_Accuracy*(y_{true}, y_{pred}) to calculate the accuracy of the Random Forest model.

IV. Observation & Conclusion

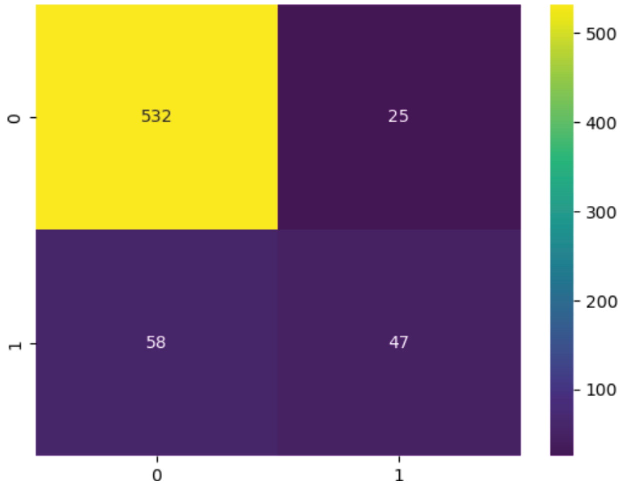


Fig. 6. Confusion Matrix

The section presents the outcomes of the classifier used for the prediction. 35 features are selected for 1470 employees, such as Age, Job satisfaction level, Salary, Overtime, Number of companies worked, and other factors. It can impact employee attrition. Random Forest, are evaluated using various binary and numeric features, using their scikit-learn [20] implementations. From the standard confusion matrix [21] as shown in Fig. 6, the performance of the model can be evaluated using various metrics such as accuracy, precision, recall and F1-score for the classifiers.

A. Observations

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

TABLE II. Result Obtained from Confusion Matrix

TP (True Positive)	532
FP (False Positive)	25
TN (True Negative)	47
FN (False Negative)	58

From confusion matrix, TABLE II is obtained. Using this data, we find accuracy equal to 0.87462, precision equal to 0.95511, recall equal to 0.90169 and F1-score equal to 0.92762.

TABLE III. Comparison Table

Performance Metrics	Performance Score of the research	Performance score of research, 2022 by Akinode John Lekan
Accuracy Score	0.87462	0.836735
Precision	0.95511	0.666667
Recall	0.90169	-
F1-Score	0.92762	0.217391

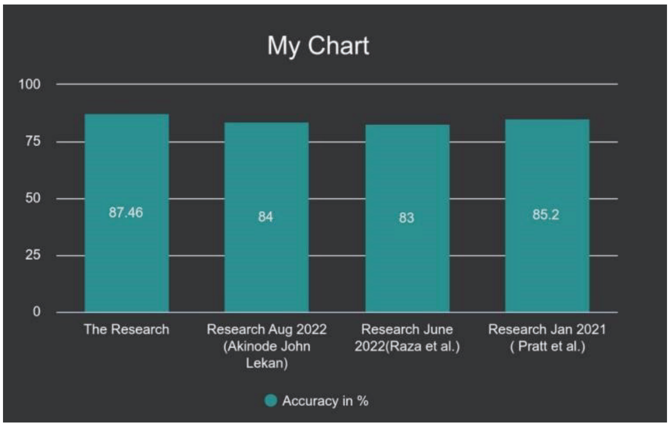


Fig. 7. Observation

Additionally, as seen in Fig. 5, Random Forest classifier developed by Sklearn contains an extremely useful attribute for determining the significance of features. This attribute displays the characteristics in our dataset that the Random Forest algorithm has determined to be most significant.

V. Conclusion

Employee attrition is a major issue that affects organizational productivity, capital, and results in the loss of valuable skills and expertise. Excessive turnover rates can affect long-term business goals, lower productivity, and have a negative impact on staff morale. In the research, random forest algorithm has been implemented on the employee dataset for prediction of employee attrition beforehand. The use of the Random Forest algorithm provides a effective tool for organizations seeking to mitigate the adverse effects of turnover [22]. From the research, it is discovered that monthlyincome, overtime, joblevel are the most significant feature that impact the employee decision as shown in Fig.

5. The research concludes that early attrition identification might help the organization to address the cause of employee attrition and put more efforts to minimize it. The results can help the human resources (HR) [23] department by providing crucial details regarding a possible employee choice to quit the company.

VI. Future Work

In future, the accuracy of the research can be increased. Also, the research can further be taken forward by comparing Random Forest algorithm with different supervised machine learning algorithm like Logistic Regression, SVM, Naive Bayes, Decision Tree and so on. Further Deep learning [24] can be used to enhance the accuracy.

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